Technical specification template: Uninterruptible Power Supply Systems

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A6. Generated 2026-11-08 18:00 UTC. Package version 0.1.0.dev0.

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Document control

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1. Scope

This template defines the technical requirements for uninterruptible power supply systems for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEC 62040 series (uninterruptible power systems), including performance classification
- IEC 61000 series (EMC and harmonics)
- IEEE 519 (harmonic limits), if adopted by the project
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit study pending utility, transformer and generator data]	NOT CALCULATED

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
UPS per power path	2	PRELIMINARY_ASSUMPTION. 2N: each UPS carries the whole critical load after loss of the other path
Rated active power (each)	2,500 kW	UNVERIFIED_PLACEHOLDER
Rated apparent power (each)	2,500 kVA	UNVERIFIED_PLACEHOLDER. Both kW and kVA limits apply
Efficiency (design point)	96 %	UNVERIFIED_PLACEHOLDER. A single point. The manufacturer shall declare the efficiency curve
Input power factor	0.99	UNVERIFIED_PLACEHOLDER
Autonomy	10 min at the design load	PRELIMINARY_ASSUMPTION
Topology, bypass, paralleling	[Project to select]	Project decision

5. Technical requirements

5.1 The UPS shall deliver its rated kW and kVA continuously at the site conditions; overload capability shall be declared but shall not be used to meet continuous loading.

5.2 The manufacturer shall declare efficiency at 25, 50, 75 and 100 per cent load [project to confirm the points].

5.3 The input current harmonic distortion, input power factor, rectifier walk-in time and generator compatibility shall be declared.

5.4 Output voltage and frequency regulation, short-circuit contribution and fault clearing capability shall be declared.

5.5 Battery interface, monitoring, alarms, communication and maintenance bypass: [project to define].

6. Vendor data schedule

The tenderer shall complete the schedule below with its offer: the value offered, its unit, and the name and revision of the document that states it. A value without a source document is not accepted. The parameters are those in the manufacturer-data register; supplied values are imported into the calculations through the data override mechanism, and a value stays NOT VERIFIED until a reviewer records an attestation against its source (name, organisation, date, evidence reference, scope).

Parameter id	Parameter	Datasheet required	Current status	Value offered	Unit	Document and revision
equipment.ups_rating	UPS rating	UPS data sheet	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
equipment.u	UPS rating	UPS data	UNVERIFIE	[vendor to	[vendor to	[vendor to

ps_rating_kva	(each), apparent power	sheet	D_PLACEHOLDER	complete]	complete]	complete]
equipment.ups_efficiency	UPS efficiency	UPS data sheet	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
equipment.ups_input_power_factor	UPS input power factor	UPS data sheet	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
derating.ups_ambient_factor	UPS ambient derating factor	Manufacturer derating curve for ambient conditions (ups)	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
derating.ups_altitude_factor	UPS altitude derating factor	Manufacturer derating curve for altitude conditions (ups)	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
derating.ups_installation_factor	UPS installation derating factor	Manufacturer derating curve for installation conditions (ups)	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
generators.ups_rectifier_walk_in_s	UPS rectifier walk-in time	UPS data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
generators.ups_input_current_thd_pct	UPS input current THD	UPS data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]

Model, manufacturer, country of manufacture, and the type-test reports for the offered design: [vendor to complete].

7. Compliance schedule

The tenderer shall state, for every standard in section 2 and every technical requirement in section 5, whether the offer complies. 'Complies' needs an evidence reference (type-test report, certificate, calculation, drawing or data sheet with revision). A statement without evidence is recorded as not demonstrated. The requirement references below are the numbered items of section 5 of this template; no clause of any standard is quoted.

Standard named in section 2	Edition offered	Complies / complies with comment / does not comply	Evidence reference	Comment or deviation
IEC 62040 series (uninterruptible power systems), including performance classification	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]
IEC 61000 series (EMC and harmonics)	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]

IEEE 519 (harmonic limits), if adopted by the project	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]
Local requirements [project to add]	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]

Requirement (section 5)	Complies / complies with comment / does not comply	Evidence reference	Comment or deviation
5.1	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.2	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.3	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.4	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.5	[vendor to complete]	[vendor to complete]	[vendor to complete]

8. Inspection and quality requirements

8.1 Review of the datasheet, single-line and control philosophy.

8.2 Inspection of enclosures and power components.

8.3 Routine tests on every unit as required by the applicable standard and this specification.

8.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

8.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

8.6 Calibration certificates for test instruments.

9. Factory acceptance test (FAT) requirements

9.1 Functional tests: transfer to and from battery and bypass, load steps, overload behaviour within declared limits, efficiency measurement at the declared points.

9.2 Input harmonics and walk-in verification [project to define].

9.3 Witnessing: [owner / independent inspector to attend].

9.4 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9.5 Acceptance criteria are taken from the applicable standard and this specification [project to define]; a test passed against a criterion not stated in advance is not accepted.

9.6 Equipment shall not be shipped until the FAT report is accepted or a concession is recorded and approved by the owner.

10. Site acceptance test (SAT) requirements

10.1 Input and output checks; transfer to and from battery and bypass; maintenance bypass operation and interlocking; alarm and communications checks; a load test with the site load or load banks as approved.

10.2 A SAT procedure shall be submitted for approval before the tests. It states the test, the method, the instrument, the acceptance criterion and the witness, with the criteria taken from the applicable standard and this specification [project to define].

10.3 SAT is carried out on the installed equipment with its final connections, after the installation inspection and before commissioning loads are applied.

10.4 Results are recorded on signed sheets. Any deviation, retest and concession is listed with its disposition. A SAT is not passed while a deviation is open.

10.5 SAT results are compared with the FAT results; a significant difference is investigated and recorded.

11. Commissioning requirements

11.1 Installation checks, battery connection checks, parameter and alarm checks.

11.2 Load bank and autonomy test together with the battery; generator interaction test.

11.3 Failure tests as part of integrated systems testing.

11.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

11.5 Integrated systems testing of the whole power chain (utility loss, generator start and transfer, UPS on battery, loss of one power path, return to normal) is carried out under a project procedure [project to define]. This equipment's commissioning is not complete until it has taken part in that test.

12. Submittals and documentation

Document	When required	Status
Completed vendor data schedule (section 6) and compliance schedule (section 7)	With the tender	[vendor to complete]
Datasheets and general arrangement drawings	With the tender, final before manufacture	[vendor to complete]
Type-test reports and certificates for the offered design	With the tender	[vendor to complete]
Inspection and test plan (ITP)	Before manufacture	[vendor to complete]
FAT procedure and report	Before and after the FAT	[vendor to complete]
Installation instructions and SAT procedure	Before delivery	[vendor to complete]
Operation and maintenance manuals, as-built drawings and final data	Before commissioning; as-built at handover	[vendor to complete]
Recommended spare parts and special tools lists	With the tender and at handover	[vendor to complete]
Training plan and materials [project to define]	Before commissioning	[vendor to complete]
Warranty terms [project to define]	With the tender	[vendor to complete]

13. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.

- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- The vendor data schedule (section 6) and the compliance schedule (section 7) are blank until a tenderer completes them.
- FAT, SAT and commissioning acceptance criteria are [project to define].
- This template has not been reviewed by an independent engineer.